THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET A

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bus covers 120 km in 2 hours. Its average speed is: 1

(a) 30 km/h **(b)** 70 km/h **(c)** 60 km/h **(d)** 240 km/h

2. A bus covers 360 km in 6 hours. Its average speed is: 1

(a) 70 km/h **(b)** 2160 km/h **(c)** 30 km/h **(d)** 60 km/h

3. A car moves at a constant 60 km/h for 3 hours. The distance covered is: 1

(a) 180 km **(b)** 240 km **(c)** 20 km **(d)** 90 km

4. A car moves at a constant 80 km/h for 1.5 hours. The distance covered is: 1

(a) 53.3333 km **(b)** 200 km **(c)** 120 km **(d)** 60 km

5. A cyclist travels 240 km at 60 km/h. The time taken is: 1

(a) 5 hours **(b)** 14400 hours **(c)** 8 hours **(d)** 4 hours

6. 108 km/h is equal to: 1

(a) 388.8 m/s **(b)** 1.8 m/s **(c)** 30 m/s **(d)** 108 m/s

7. 5 m/s is equal to: 1

(a) 18 km/h **(b)** 300 km/h **(c)** 1.4 km/h **(d)** 50 km/h

8. The time period of a simple pendulum depends mainly on its: 1

(a) colour **(b)** material of bob **(c)** mass of bob **(d)** length

9. A water clock is also called: 1

(a) hourglass **(b)** sundial **(c)** stopwatch **(d)** clepsydra

10. An object covering equal distances in equal intervals of time has: 1

(a) non-uniform motion **(b)** variable speed

(c) zero speed **(d)** uniform motion

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. Define time period of a pendulum. 2

12. What is the SI unit of time and speed? 2

13. Define speed. Write its formula. 2

14. Differentiate between uniform and non-uniform motion. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A train covers 300 km in 5 hours. Find its speed in km/h and m/s. 3

16. How long will a bus take to cover 210 km at 42 km/h? 3

17. Describe an activity to find the time period of a simple pendulum. 3

OR

Explain how a sundial works. 3

18. A man walks at 5 km/h for 2 hours and cycles at 15 km/h for 1 hour. Find the total distance and average speed. 3

OR

The distance–time data: 0 s 0 m, 2 s 10 m, 4 s 20 m, 6 s 30 m. Is the motion uniform? Find speed. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Define speed and give its SI unit.
(ii) A car travels 180 km in 3 hours. Find its speed.
(iii) Express it in m/s. 5

OR

Case study: Ravi times a pendulum: 20 oscillations take 36 s. 5
(i) Find time period.
(ii) How many oscillations in 1 min?
(iii) What if the string is longer?
(iv) Why count 20 oscillations?

20. (i) Describe the working of a sundial and clepsydra.
(ii) State advantages and limitations of each. 5

OR

Case study: A bus covers 60 km in the first hour, 40 km in the second hour and 80 km in the third. 5
(i) Total distance.
(ii) Total time.
(iii) Average speed.
(iv) Is the motion uniform?

END OF QUESTION PAPER — Total: 40 Marks

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